MILITARY SPECIFICATION

FUZE, PIBD, XM579

1. SCOPE

1.1 This specification covers the metal parts, loading, assembling and packing for one type of fuze designated as Fuze, PIBD, XM579.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-D-14978	-	Detonator, Stab M55
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

FEDERAL

FED-STD-151	-	Metals; Test Methods

FSC: 1390

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environment and Performance Tests for
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. Army Munitions Command

9226650	-	Fuze, PIBD, XM579
9234336	-	Box, Fiber, Inner, for Fuze, PIBD, XM579 for
9234337	-	Box, Fiber, Outer, for Fuze, PIBD, XM579 for 30MM Cartridge

PUBLICATIONS

ARMY

ET-9226650	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 171-179 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg.) F-9226650, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Jolt and Jumble.-The fuzes, when tested as specified in 4.4, shall withstand these tests without becoming hazardous to anyone handling or using them.

3.4 Transportation vibration.-The fuze shall be safe and operable after being tested as specified in 4.4.5.

3.5 Waterproofness.-A continuous stream of bubbles shall not emerge from the fuze when tested as specified in 4.4.7.

3.6 Non-functioning.-The fuze shall not be armed within 74 feet of the gun muzzle when tested as specified in 4.4.9.

3.7 Functioning.-The fuze shall be armed and shall function high order when tested as specified in 4.4.8.

3.8 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

3.9 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.5).

3.10 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1). The first article sample shall consist of the following:

a. Two hundred fifty nine (259) assembled fuzes
b. Fifty (50) of each subassembly
c. Twenty-five (25) of each piece part

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Detonators of one lot interfix number from one manufacturer.
b. Metal and plastic parts of one lot interfix number from one manufacturer.
c. HMX from one lot number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.2.1 Weight, rotor (See Dwg. 9226663 covering a detail of Dwg. 9226660).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Weight improper	Balance	01001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	01002

4.3.2.2 Rotor (See Dwg. 9226662 covering a detail of Dwg. 9226660).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Profile	Gage	02001
102.	Thickness	Gage	02002
103.	Excessive voids	Visual	02003
104.	Presence of required bosses, cavities, holes, as required	Visual	02004
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	02005

4.3.2.3 Shaft, rotor (See Dwg. 9226664 covering a detail of Dwg. 9226660).

Category	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Length from tapered end to opposite end	Gage	03001
102.	Smallest outside diameter	Gage	03002
103.	Largest outside diameter	Gage	03003
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	03004

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.2.4 Housing, rotor (See Dwg. 9226658 covering a detail of Dwg. 9226657).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.50 Percent		
101.	Diameter of large cavity	Gage	04001
102.	Diameter of small cavity	Gage	04002
103.	Concentricity of small cavity with large cavity	Gage	04003
104.	Depth of large cavity	Gage	04004
105.	Depth of small cavity	Gage	04005
106.	Diameter of small hole	Gage	04006
107.	Finish missing or incorrect	Visual	04007
108.	Depth to bend for indicated length	Gage	04008
109.	Distance to end of tab from small hole	Gage	04009
110.	Diameter of large hole	Gage	04010
111.	Location of small hole	Gage	04011
112.	True position of cutout	Gage	04012
113.	Width of cutout	Gage	04013
Minor	AQL 2.50 Percent		
201.	Symmetry of key	Gage	04014
202.	Symmetry of tab	Gage	04015
203.	Width of key	Gage	04016
204.	Evidence of poor workmanship (See 3.8)	Visual	04017
205.	Thickness of wall	Gage	04018

4.3.2.5 Hub, delay ribbon (See 9226659 covering a detail of Dwg. 9226657).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Total length	Gage	05001
102.	Length of largest outside diameter	Gage	05002
103.	Length of smallest outside diameter	Gage	05003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

104.	Small inside diameter	Gage	05004
105.	Large inside diameter	Gage	05005
106.	Concentricity of large inside diameter with largest outside diameter	Gage	05006
107.	Concentricity of small inside diameter with second largest outside diameter	Gage	05007
108.	Second largest outside diameter	Gage	05008
109.	Depth of cavity	Gage	05009
110.	Largest outside diameter	Gage	05010
Minor	AQL 1.50 Percent		
201.	Width of keyway (2 places)	Gage	05011
202.	Symmetry of keyway to second largest outside diameter	Gage	05012
203.	True position of keyways	Gage	05013
204.	Evidence of poor workmanship (See 3.8)	Visual	05014

4.3.2.6 Pin, detent (See Dwg. 9226653 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Burrs	Visual-Manual	06001
102.	Diameter	Gage	06002
103.	Length	Gage	06003
104.	Protective finish missing or bare spot	Visual	06004
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	06005

4.3.2.7 Spring, compression (See Dwg. 9226655).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Load at compressed lengths	Gage	07001
102.	Solid height, max.	Gage	07002

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

Minor	AQL 1.0 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	07003
202.	Inside diameter	Gage	07004

4.3.2.8 Liner, delay ribbon, inner (See Dwg. 9226670 covering a detail of Dwg. 9226668).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.0 Percent		
101.	Length in direction of skiving	Gage	08001
102.	Angle at ends	Gage	08002
103.	Parallelism	Gage	08003
104.	Burrs or feathered edges	Visual	08004
105.	Small angle	Gage	08005
Minor	AQL 1.50 Percent		
201.	Width	Gage	08006
202.	Thickness	Gage	08007
203.	Evidence of poor workmanship (See 3.8)	Visual	08008

4.3.2.9 Liner, delay ribbon, outer (See Dwg. 9226683 covering a detail of Dwg. 9226668).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Length perpendicular to parallel edges	Gage	09001
102.	Angle at ends	Gage	09002
103.	Parallelism	Gage	09003
104.	Burrs or feathered edges	Visual	09004
105.	Straightness	Gage	09005
Minor	AQL 1.50 Percent		
201.	Thickness	Gage	09006
202.	Width	Gage	09007
203.	Evidence of poor workmanship (See 3.8)	Visual	09008

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.2.10 Liner, delay ribbon, middle (See Dwg. 9226671 covering a detail of Dwg. 9226668).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Length perpendicular to parallel edge	Gage	10001
102.	Angle at ends	Gage	10002
103.	Parallelism	Gage	10003
104.	Straightness	Gage	10004
105.	Burrs or feathered edges	Visual	10005
Minor	AQL 1.50 Percent		
201.	Thickness	Gage	10006
202.	Width	Gage	10007
203.	Evidence of poor workmanship (See 3.8)	Visual	10008

4.3.2.11 Ribbon, delay (See Dwg. 9226669 covering a detail of Dwg. 9226668).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Ribbon damaged or improperly wound	Visual	11001

4.3.2.12 Shell, ogive (See Dwg. 9226673 covering a detail of Dwg. 9226672).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Location of large basic diameter	Gage	12001
102.	Location of small basic diameter	Gage	12002
103.	Diameter of internal shoulder	Gage	12003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

104.	Concentricity of larger internal shoulder to small basic diameter	Gage	12004
105.	Location of larger internal shoulder from nose end	Gage	12005
Minor	AQL 2.50 Percent		
201.	Total length	Gage	12006
202.	Wall thickness at nose end	Gage	12007
203.	Wall thickness for indicated length	Gage	12008
204.	Presence of smaller internal shoulder	Visual	12009
205.	Evidence of poor workmanship (See 3.8)	Visual	12010

4.3.2.13 Button, ogive (See Dwg. 9226674 covering a detail of Dwg. 9226672).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Profile	Gage	13001
102.	Presence of cavity	Visual	13002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	13003

4.3.2.14 Pin, firing (See Dwg. 9226666 covering a detail of Dwg. 9226665).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Diameter of fluted end, max	Gage	14001
102.	Concentricity of flange diameter with diameter of fluted end	Gage	14002
103.	Distance from point to end of flange	Gage	14003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

104.	Profile of point	Gage	14004
105.	Bulge diameter	Gage	14005
106.	Concentricity of bulge with large diameter of flutes	Gage	14006
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	14007

4.3.2.15 Housing, firing pin (See Dwg. 9226667 covering a detail of Dwg. 9226665).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Profile	Gage	15001
102.	Diameter of hole	Gage	15002
103.	Perpendicularity of hole to open end	Gage	15003
104.	Concentricity of outside diameter to hole	Gage	15004
105.	Concentricity of second largest outside diameter to hole	Gage	15005
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	15006

4.3.2.16 Plate, rotor housing (See Dwg. 9226652 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Outside diameter	Gage	16001
102.	Concentricity of inside diameter with outside diameter	Gage	16002
103.	Parallelism of faces	Gage	16003
104.	Diameter of small hole	Gage	16004

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

105.	Straightness of faces	Gage	16005
106.	Width of neck of radially bent mid-section	Gage	16006
107.	Thickness of neck of radially bent midsection	Gage	16007
108.	Location of bent midsection from outside diameter	Gage	16008
109.	True position of small hole	Gage	16009
110.	Width of tab	Gage	16010
111.	True position of tab	Gage	16011
Minor	AQL 1.00 Percent		
201.	Inside diameter of cutout	Gage	16012
202.	Evidence of poor workmanship (See 3.8)	Visual	16013

4.3.2.17 Housing, lower (See Dwg. 9226651 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.50 Percent		
101.	Pitch diameter of external thread	Gage	17001
102.	Major diameter of external thread	Gage	17002
103.	Pitch diameter of internal thread	Gage	17003
104.	Minor diameter of internal thread, max	Gage	17004
105.	Concentricity of minor diameter of internal thread with pitch diameter of external thread	Gage	17005
106.	Length from bottom of cavity to opposite end	Gage	17006
107.	Knurl missing or improper	Visual	17007
108.	Diameter of cavity	Gage	17008
109.	Concentricity of cavity with pitch diameter of external thread	Gage	17009
110.	Parallelism	Gage	17010
111.	Largest outside diameter	Gage	17011
112.	Second smallest outside diameter	Gage	17012

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

113.	Concentricity of largest outside diameter to pitch diameter of external thread	Gage	17013
114.	Concentricity of second smallest outside diameter to pitch diameter of external thread	Gage	17014
inor	AQL 1.50 Percent		
201.	Smallest outside diameter	Gage	17015
202.	Evidence of poor workmanship (See 3.8)	Visual	17016
203.	Total length	Gage	17017
204.	Effective length of internal thread, min	Gage	17018

4.3.2.18 Housing, upper (See Dwg. 9226654 covering a detail of Dwg. 9225650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Outside diameter	Gage	18001
102.	Inside diameter of large cavity	Gage	18002
103.	Concentricity of inside diameter of large cavity to outside diameter	Gage	18003
104.	Inside diameter of small cavity	Gage	18004
105.	Width of raceways in small cavity (3 places)	Gage	18005
106.	Length from bottom of small cavity to open end of large cavity	Gage	18006
107.	Concentricity of inside diameter of small cavity to outside diameter	Gage	18007
108.	Parallelism	Gage	18008
109.	Diameter of hole	Gage	18009
110.	Depth of large cavity	Gage	18010
111.	Symmetry of raceways to inside diameter	Gage	18011

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

Category		Method of Inspection	Defect Code
Minor	AQL 1.00 Percent		
201.	Protective finish improper	Visual	18012
202.	Evidence of poor workmanship (See 3.8)	Visual	18013

4.3.2.19 Cup, lead (See Dwg. 9226677 covering a detail of Dwg. 9226675).

Category		Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Outside diameter	Gage	19001
102.	Inside diameter	Gage	19002
103.	Thickness at closed end	Gage	19003
Minor	AQL 0.65 Percent		
201.	Total length	Gage	19004
202.	Evidence of poor workmanship (See 3.8)	Visual	19005

4.3.2.20 Spitback, adapter (See Dwg. 9226676 covering a detail of Dwg. 9226675).

Category		Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Length from open end to flange	Gage	20001
102.	Diameter of cavity	Gage	20002
103.	Depth of cavity	Gage	20003
104.	Diameter of hole	Gage	20004
105.	Pitch diameter	Gage	20005
106.	Concentricity of cavity diameter to pitch diameter	Gage	20006
107.	Concentricity of hole to pitch diameter	Gage	20007
108.	Perpendicularity of pitch diameter to flange	Gage	20008
Minor	AQL 1.00 Percent		
201.	Distance between faces of hexagon	Gage	20009
202.	Evidence of poor workmanship (See 3.8)	Visual	20010

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.2.21 Cup, acceptor (See Dwg. 9226679 covering a detail of Dwg. 9226675).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Outside diameter of cavity	Gage	21001
102.	Diameter of hole	Gage	21002
103.	Flange diameter	Gage	21003
104.	Thickness of wall	Gage	21004
105.	Concentricity of hole with outside diameter of cavity	Gage	21005
106.	Concentricity of flange diameter with outside diameter of cavity	Gage	21006
Minor	AQL 1.00 Percent		
201.	Total length	Gage	21007
202.	Evidence of poor workmanship (See 3.8)	Visual	21008

4.3.2.22 Disc (See Dwg. 9226680 covering a detail of Dwg. 9226675).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major:	None defined		
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	22001

4.3.2.23 Sleeve (See Dwg. 9226678 covering a detail of Dwg. 9226675).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Length	Gage	23001

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

Minor	AQL 1.50 Percent		
201.	Outside diameter	Gage	23002
202.	Thickness of wall	Gage	23003
203.	Evidence of poor workmanship (See 3.8)	Visual	23004

4.3.2.24 Spitback lead assembly (See Dwg. 9226675 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Length from flange to end	Gage	24001
102.	Component missing or improperly assembled	Visual	24002
Minor	AQL 1.00 Percent		
201.	Threads damaged	Visual	24003
202.	Evidence of poor workmanship (See 3.8)	Visual	24004

4.3.2.25 Firing pin assembly (See Dwg. 9226665 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Firing pin missing, damaged or not properly assembled	Visual-Manual	25001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	25002

4.3.2.26 Rotor assembly (See Dwg. 9226660 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

Category	Defect	Method of Inspection	Defect Code
Major	AQL 0.65 Percent		
101.	Components missing or improperly assembled	Visual-Manual	26001
102.	Rotor weight above flush (2 places)	Gage	26002
103.	Rotor shaft above flush	Gage	26003
104.	Distance from rear boss to rotor shaft, max	Gage	26004
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	26005

4.3.2.27 Rotor housing and hub assembly (See Dwg. 9226657 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Hub improperly assembled	Visual-Manual	27001
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	27002

4.3.2.28 Delay ribbon assembly (See Dwg. 9226668 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical			
1.	Delay ribbon loose	Visual	28001
2.	Liners missing	Visual	28002
Major	AQL 0.40 Percent		
101.	Liners improperly assembled	Visual	28003

4.3.2.29 Shell and button assembly (See Dwg. 9226672 covering a detail of Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

Major	AQL 0.40 Percent		
101.	Distance from nose to button, max	Gage	29001
102.	Dimples missing or improper	Visual	29002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	29003

4.3.2.30 Fuze PIBD XM579 prior to assembly of Spitback (See Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical			
1.	Rotor armed or partially armed	Visual	30001
Major	AQL 0.40 Percent		
101.	Component missing, damaged or improperly assembled	Visual	30002
102.	Length nose to shoulder	Gage	30003
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.8)	Visual	30004

4.3.2.31 Fuze, PIBD XM579 after assembly of Spitback (See Dwg. 9226650).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Spitback assembly not fully seated	Visual	31001
Minor	AQL 1.00 Percent		
201.	External threads damaged	Visual	31002
202.	Evidence of poor workmanship (See 3.8)	Visual	31003

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.2.32 Box, fibre, prior to sealing (See Dwg. 9234336)

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Incorrect number of trays	Visual	32001
102.	Incorrect number of fuzes with spitback	Visual	32002
103.	Fuze inverted	Visual	32003
Minor	AQL 1.50 Percent		
201.	Empty tray missing on top of exposed fuzes	Visual	32004
202.	Fillers missing (when applicable)	Visual	32005
203.	Liner missing	Visual	32006

4.3.2.33 Box, fibre (See Dwg. 9234337).

Category	Defect	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Contents loose	Manual	33001
102.	Box damaged	Visual	33002
103.	Improper assembly	Visual	33003
Minor	AQL 0.65 Percent		
201.	Marking misleading or illegible	Visual	33004

4.3.3 Testing

4.3.3.1 Dimensional control of die cast parts.-In place of the normal sampling associated with the Classification of Defects and with the approval of the Contracting Officer, a sample of at least three (3) parts (as cast) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

4.3.3.2 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.3 Dimensional control of plastic molded parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved by the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

Contracting Officer (See Note 1), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 5000 parts or at the end of each shift, providing the number of parts produced during that shift is less than 5000 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

(NOTE 1) - In establishing a curing time, dimensionally check three (3) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained.. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, SMUPA-ND3. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.3.3.4 Check test for deterioration of detonators (See MIL-D-14978) Major defect (Code No. 34001).-If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

since previous tests, the detonator lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-D-14978 immediately before the detonator lot is assembled into the fuzes. This test shall be performed by the contractor on detonators selected by the Government Inspector at the facility assembling the detonator into the fuze. (See 6.4).

This test will be conducted at Government expense without cost to the contractor who loaded the detonator or the contractor assembling the detonator into the fuze, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling in storage, etc., permitted while the detonator lot was under the responsibility of either contractor.

The Contracting Officer shall arrange to have the contractor reimbursed for the expenses incurred in the performance of check tests.

4.3.3.5 Push-out test of detonator on rotor assembly (See Dwg. 9226660) Major defect, Code No. 35001.-The sampling plan for this test shall be in accordance with Military Standard MIL-STD-105, Code Letter K, single plan with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.6 Push-out test of firing pin on firing pin assembly (See Dwg. 9226665) Major defect, Code No. 36001.-The sampling plan for this test shall be in accordance with Military Standard MIL-STD-105, Code Letter K, single plan with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.7 Push-out test of button on shell and button assembly (See Dwg. 9226672) Major defect, Defect Code No. 37001.-The sampling plan for this test shall be in accordance with Military Standard MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

4.3.3.8 Jolt (See para. 3.3) Major defect, Defect Code No. 38001.-Twelve (12) fuzes shall be selected from each lot and subjected to this test. If any fuze assembly fails to comply with the applicable requirement, the lot shall be rejected. This test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.3.9 Jumble (See para. 3.3), Major defect, Defect Code No. 39001.-Twelve (12) fuzes shall be selected from each lot and subjected to this test. If any fuze assembly fails to comply with the applicable requirements, the lot shall be rejected. This test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

4.3.3.10 Transportation-Vibration (see para. 3.4), Major defect, Defect Code No. 40001.-One hundred seventy-two (172) fuzes shall be selected from each lot, placed in their regular shipping containers and subjected to this treatment. The container shall be treated as specified in 4.4.5, using equipment in accordance with 4.3.4.

4.3.3.11 Torque of fuze assembly, Major defect, Defect Code No. 41001.-After completion of treatment specified in 4.3.3.10, one hundred sixty (160) fuzes shall be selected from the sample and subjected to this test. If six (6) or more fuze assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. Assemblies which fail this test shall be discarded and replaced with units which had been subjected to test specified in 4.3.3.10, prior to subsequent testing. This test shall be performed as specified in 4.4.6, using equipment in accordance with 4.3.4.

4.3.3.12 Waterprodfness (See para. 3.5), Major defect, Defect Code No. 42001.-After completion of tests specified in 4.3.3.10 and 4.3.3.11, one hundred sixty of the sample fuze assemblies shall be subjected to this test. If six (6) or more fuze assemblies fail to comply with the applicable requirement, the lot shall be rejected. Assemblies which fail this test shall be discarded and replaced with units which had been subjected to treatment specified in 4.3.3.11, prior to subsequent testing. This test shall be performed as specified in 4.4.7, using equipment in accordance with 4.3.4.

4.3.3.13 Functioning (See 3.7) Major defect, Defect Code No. 43001.-After completion of tests specified in 4.3.3.10, 4.3.3.11 and/or 4.3.3.12, one hundred sixty (160) fuze assemblies shall be subjected to this test. Eighty (80) rounds shall be fired to impact at approximately 2500 meters from the launcher on a pea gravel target. The remaining eighty rounds shall be fired to impact on 16 ga (.063) steel sheet at 250 feet from the launcher. If the combined total of fuzes not functioning exceeds four (4) or if three (3) or more do not function in either individual test of eighty (80), the lot shall be rejected. This test shall be performed as specified in 4.4.8, using equipment in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.3.14 Non-functioning (See para. 3.6) Critical Defect, Defect Code No. 44001.-Twenty (20) additional fuze assemblies shall be selected from each lot and subjected to this test. If any fuze assembly functions upon impact with steel sheet at seventy four (74) feet from the point of launching, the lot shall be rejected. This test shall be performed as specified in 4.4.9, using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9226650 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Push-out test of detonator from rotor assembly.-The rotor assembly shall be placed in a fixture as shown on the Equipment List and the force specified shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.2 Push-out test of firing pin of firing pin assembly The firing pin assembly shall be placed in a fixture as shown on the Equipment List and the force specified on the applicabl drawing shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.3 Push-out test of button from shell and button assembly.-The shell and button assembly shall be placed on a fixture as shown on the Equipment List and the force specified on the applicable drawing shall be applied. This test is a destructive test. Parts so tested shall not be returned to th lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.4.4 Jolt and Jumble.-The fuzes with spitbacks assembled shall be tested in accordance with the procedures specified in Military Standard MIL-STD-331. These tests are destructive tests. Parts so tested shall not be returned to the lot.

4.4.5 Transportation-Vibration.-The fuzes with spitbacks assembled, packed in their regular shipping containers, shall be tested in accordance with Military Standard MIL-STD-331. Fuzes not expended in subsequent testing shall not be returned to the lot.

4.4.6 Torque of fuze assembly.-The fuze shall be held firmly and the torque specified on the applicable drawing shall be applied. Fuzes so tested shall be used for the test specified in 4.3.3.12.

4.4.7 Waterproofness.-The fuzes shall be assembled to a simulated body as specified on the Equipment List, and then immersed in water to a depth of not less than three (3) inches. The chamber shall be subjected to a vacuum of three (3) psi gage min., for a period of ten (10) seconds, min. The water shall contain the wetting agent Nacconal NR (sodium alkyl aryl sulfonate) or equal, in a concentration of approximately 1½ teaspoonful per gallon of water. Fuze assemblies so tested shall be used for the test specified in Paragraph 4.3.3.12.

4.4.8 Functioning.-The fuze assembly shall be assembled to a loaded XM553 body assembly, then to a loaded cartridge case, XM193. The cartridge shall be fired to impact at approximately 2500 meters from the launcher. Other cartridges shall be fired to impact on a vertical target of 16 ga (.063) steel sheet. Proper functioning shall be assumed if the round explodes high order on impact with the target.

4.4.9 Non-functioning.-The fuze assembly shall be assembled to a loaded XM553 body assembly, then to a loaded cartridge case, XM193. The cartridge shall be fired to impact on a vertical target of 16 ga (.063) steel sheet, placed at a distance of seventy-four (74) feet from the launcher. Observation shall be made for non-arming as evidenced by non-functioning upon impact on the target.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The fuzes shall be preserved and packaged in accordance with Dwg. 9234336.

5.2 Packing.

5.2.1 Level C.-The fuzes shall be packed in accordance with Dwg. 9234337.

5.3 Marking.-Marking shall be in accordance with Dwg. 9234336, 9234337 and Department of Transportation Regulation 49 CFR 171-178.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND-13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonator assemblies. The tests shall be conducted at Government expense without cost to the contractor who

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

loaded the detonator assemblies, or to the contractor assembling the detonator assemblies into the fuzes; and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator assembly lot was under the jurisdiction of either contractor.

6.5 Submission of results of contractor-conducted examinations and tests.- Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number:
1390-A826

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.